ECON 2123

Macroeconomics Integrated Study Notes

Lectures 1-9 and Tutorials 1-3

English explanations with Chinese support for difficult ideas

Measurement · Goods market · Money and bonds · IS-LM

These notes connect measurement, the goods market, financial markets, and the IS-LM model into one framework. They cover the academic content in all five supplied lecture decks and all three tutorial PDFs, including numerical exercises and questions left open in the slides. Course administration, repeated outlines, and decorative news illustrations are condensed. Later topics merely announced in the slides, such as the full AS-AD model, open-economy models, and growth theory, are outside this set of materials.

The lecture slides determine the model used here. In particular, the course uses an **upward-sloping LM curve with a given money supply**. The horizontal LM curve in recent Blanchard editions describes a different policy assumption. Additional explanations and practice examples are labelled when they extend the supplied material. Historical charts are discussed as course illustrations, not as updated economic data.

How to study: Read Chapters 1-9 in order once. Then redo the worked examples without looking at the answers, draw the graphs from their economic mechanisms, and use Chapter 10 for retrieval practice. Appendix A maps every source to these notes. Appendix B records corrections and clarifications to the original materials.

Prepared from the supplied materials | 28 September 2026

Contents

1. The common logic behind the course 3

2. GDP: production, income, and expenditure 6

3. Real GDP, chain indexes, and inflation 9

4. Employment, unemployment, and participation 13

5. The goods market and the Keynesian cross 14

6. Saving, fiscal policy, and stabilisers 17

7. Money, bonds, and monetary policy 21

8. Building the IS-LM model 25

9. Policy effects, feedback, and policy mixes 28

10. Revision sheet and self-test 33

Appendix A. Source and coverage map 37

Appendix B. Corrections and interpretation notes 38

Reading route

Chapters 1-4 establish the language and measurement. Chapters 5-7 develop the two markets. Chapters 8-9 combine them and analyse policy. Chapter 10 provides formulas and retrieval practice. The appendices locate the sources and correct the original typos.

Worked tutorial questions are integrated beside the concepts they practise. Source references appear at the beginning of each chapter; a question-by-question map is in Appendix A.

1. The common logic behind the course

Sources: L1, slides 22-40; L2-3, slides 2-6; T2, pages 3-8; L8-9, slides 6-8.

1.1 What macroeconomics studies

Macroeconomics studies the economy as a whole: how much it produces, how many people work, how prices change, and how policy affects these outcomes. Individual spending and saving decisions interact. One person's expenditure becomes another person's income, so an action that is sensible for one household can have a different effect when everyone takes it at once.

Economic growth concerns the long-run rise in productive capacity and real income per person. **Business cycles** concern expansions and contractions around the economy's longer-run path. Real GDP per capita equals real GDP divided by population. It is useful for comparing average production per person, although it does not reveal the distribution of income or capture every aspect of welfare.

The course introduces four sets of questions. What drives growth and differences between countries? How do central banks influence interest rates and activity? How do government purchases and taxes affect demand? How do exchange rates and trade connect economies? The supplied lectures develop the closed-economy, short-run part of that agenda.

The Great Depression motivates both measurement and modelling. Policymakers need reliable aggregate statistics to identify the scale of a downturn. Keynes's demand-based analysis explains why weak expenditure can leave an economy producing below its capacity. Hicks and Hansen developed the IS-LM framework to analyse goods and financial markets together.

1.2 The chain linking the chapters

1. **Measurement:** GDP measures production, income, and expenditure from different angles. Nominal and real measures separate prices from quantities.
2. **Goods market:** Planned spending influences production. Production generates income, and income influences consumption. This feedback creates a multiplier.
3. **Financial markets:** Transactions create demand for money. The interest rate adjusts to reconcile money demand with money supply.
4. **IS-LM:** Interest rates affect investment and output; output affects money demand and interest rates. Equilibrium must satisfy both markets simultaneously.
5. **Policy:** Fiscal policy changes goods demand directly. Monetary policy changes financial conditions and then investment. Their interaction determines the final outcome.

> 中文理解**:** 生产形成收入，收入影响消费；交易需要货币，利率影响投资。IS-LM 把这两组反馈放在一起，所以不能只看政策的第一步效果。

1.3 Three kinds of equations

Kind	Meaning	Example	What it does not establish by itself
Accounting identity	True by definition when measured consistently	$Y \equiv C + I + G + NX$, with inventory investment included in I	Which variable causes another to change
Behavioural equation	Describes an assumed response	$C = c_0 + c_1(Y - T)$	An accounting truth valid under all circumstances
Equilibrium condition	A condition that jointly determined variables must satisfy	$Y = Z$; $M^s = M^d$	That the economy starts in equilibrium or adjusts instantly

Actual expenditure and planned expenditure must be distinguished. Unsold output counts as inventory investment, so actual GDP accounting still balances. The condition Y = Z in the Keynesian cross says firms produce exactly what buyers plan to buy, with no unintended inventory accumulation or depletion.

> 中文理解: “恒等式”是记账关系；“行为方程”是对人的反应的假设；“均衡条件”是模型要求供需一致的条件。三者用途不同。

1.4 Assumptions are part of the answer

The basic Keynesian cross assumes a closed economy, a fixed price level, fixed investment, and given government purchases and net taxes. Firms can respond to changes in demand by changing production. The simple consumption function has $0 < c_1 < 1$.

The financial-market model initially treats nominal income as given, assumes two assets (money and a one-year risk-free bond), and lets the central bank set money supply. Money earns zero nominal interest in this model.

IS-LM makes output and interest rates jointly endogenous. It replaces fixed investment with I(Y, i), while retaining a fixed price level and, for the standard course diagrams, a given nominal money supply. In the short run, “fixed prices” is a modelling approximation motivated by price stickiness, not a claim that every actual price is constant.

An **exogenous variable** is taken as given in the model. An **endogenous variable** is determined inside it. Investment is exogenous in the first goods-market model but endogenous in IS-LM. A bar, as in $\bar{I}$, indicates a given value.

1.5 Notation and units

Symbol	Meaning	Important distinction
Y	Real output or real income	In the model, measured consistently in goods or constant-price units
PY or $Y	Nominal GDP or nominal income	The slides' dollar sign marks a nominal value
P	Aggregate price level	A normalised index, not the price of one particular good
Z	Planned demand for goods	Equals output at goods-market equilibrium
C, I, G	Consumption, investment, government purchases	Flows per period
T	Net taxes: taxes less transfers	A rise in transfers lowers T in the simple model
Y^D	Disposable income, Y − T	Income available for consumption or saving
S	Private saving	National saving is S + T − G
M, M/P	Nominal and real money balances	Stocks at a point in time
i	Nominal interest rate	Use decimal form in calculations: 5% = 0.05
P^B, F	Bond price and face value	Different from the aggregate price level P
N, U, L	Employment, unemployment, labour force	L = N + U; unrelated to the function L(i)
u, π	Unemployment rate, inflation rate	Ratios, usually reported as percentages

Income and saving are flows. Wealth, money holdings, and government debt are stocks. A government deficit is a flow that contributes to the accumulation of debt. Buying a financial asset changes a portfolio; it is not automatically new investment expenditure in GDP.

1.6 The mathematics used in the tutorials

For a straight line y = a + bx, a is the vertical intercept and b = Δy/Δx is the slope. A larger absolute slope means a steeper line, given the same axes and scales. The lines y = x, y = 0.5x, y = 4x, y = −x, and y = −2x have slopes 1, 0.5, 4, −1, and −2 respectively. A slope describes the change in the vertical variable per unit of the horizontal variable.

A derivative gives the slope at a point. The power rule is:

> **Equation:** If $y = ax^n$, then $dy/dx = anx^{n-1}$.

Tutorial answers: $d(2x^2)/dx = 4x$; $d(0.1x)/dx = 0.1$; $d(100)/dx = 0$. Also, $d(2x - 1)/dx = 2$ and $d(2x^2 + 3x + 4)/dx = 4x + 3$.

A **partial derivative** changes one input while holding the others constant. For $y = 0.8(x_1 + 100 + 2x_2)$, $\partial y/\partial x_1 = 0.8$ and $\partial y/\partial x_2 = 1.6$. This “holding other things constant” reasoning is essential when separating the output effect and interest-rate effect on investment.

For any positive level X, its growth rate is $g_t = (X_t - X_{t-1})/X_{t-1}$. A change from 100 to 103 is 3%; a change from 200 to 198 is −1%. If an interest rate changes from 5% to 4%, it falls by **1 percentage point**, or **20% of its previous value**. These are different statements.

2. GDP: production, income, and expenditure

Sources: L2-3, slides 7-23 and 31-32; L4-5, slides 5-14; T1, pages 4-14.

2.1 What GDP counts

Gross domestic product is the value of final goods and services produced within an economy during a given period. Each part of the definition matters.

Value: Prices allow different products, such as haircuts and computers, to be added in common units.

Final: Count each product's value once. An intermediate input is used up or transformed in producing another good. Final uses include consumption and investment. A newly produced machine is a final investment good even though a firm uses it to make other goods over several years. The classification depends on use: an apple eaten by a household is final consumption; an apple processed into juice is an intermediate input.

Goods and services: Both count. Government services without market prices are generally valued through production costs.

Produced during the period: Reselling an existing car or house does not create that asset again. Newly provided dealer or brokerage services can still contribute to current GDP.

Domestic: Location of production matters. A foreign-owned factory's domestic production counts; production abroad by a domestically owned business does not.

Gross: Depreciation has not been deducted. The lectures do not develop depreciation calculations.

2.2 Three equivalent measurements

Final-output method: Sum the value of domestically produced final goods and services.

Value-added method: For each producer, subtract purchased intermediate inputs from the value of production. Add value added across domestic producers.

> **Equation:** Value added = value of output − value of intermediate inputs.

Income method: Add the incomes generated by production. In the simplified course examples, this is wages plus profits. More detailed national accounts contain additional income categories and accounting adjustments; use the simplified definitions when solving the two-firm exercises.

Wages are part of value added, so do not subtract them when calculating it. Profit subtracts both intermediate-input costs and wages from revenue in these examples. In T1's bread example, value added is 1 for the farmer, 3 − 1 = 2 for the miller, and 7 − 3 = 4 for the baker: a total of 7, the bread's final value.

> 中文理解: 增加值不是利润。增加值先扣中间投入，再分给劳动者（工资）和资本所有者（利润等）。把工资也扣掉，就不能再把结果当作增加值。

Lecture example: steel and cars

The steel firm sells steel for 100, pays wages of 80, and earns profit of 20. The car firm sells cars for 200, buys steel for 100, pays wages of 70, and earns profit of 30.

Method	Calculation	GDP
Final goods	Cars sold to final buyers	200
Value added	Steel: 100; cars: 200 − 100	200
Income	Wages: 80 + 70; profits: 20 + 30	200

Adding both firms' revenue, 100 + 200, double-counts the steel. Labour's income share is 150/200 = 75%; the profit share is 25%. The lecture's historical labour-share chart illustrates that the division of income between labour and capital can change even when total GDP rises.

If the steel firm is abroad, the car firm's domestic value added is still 200 − 100 = 100, but the steel producer's value added is foreign. Domestic GDP is therefore 100. The expenditure calculation gives the same result: final car expenditure of 200 less imported steel of 100.

Tutorial 1, Exercise 1: apples and juice

Firm A grows 200 kg of apples at 2 per kg. It sells 50 kg to households and 150 kg to Firm B. A's **total wage bill** is 120. Firm B produces 500 bottles of juice at 5 each, and pays **each of 10 workers** 200. Its wage bill is therefore 2,000.

Item	Firm A	Firm B
Revenue	200 × 2 = 400	500 × 5 = 2,500
Intermediate inputs	0	150 × 2 = 300
Value added	400	2,200
Wages	120	2,000
Profit	400 − 120 = 280	2,500 − 300 − 2,000 = 200

Final-output GDP = 50 × 2 + 500 × 5 = **2,600**. Value-added GDP = 400 + 2,200 = **2,600**. Income GDP = 120 + 2,000 + 280 + 200 = **2,600**. All three match.

The wording of the wages matters: 120 is A's total, while 200 is B's payment per worker. The tutorial uses different year labels on pages 8 and 9 for the same example; the numerical solution is unchanged.

Tutorial 1, Exercise 2: missing values

The steel company has wage costs of 80 and profit of 40, with no other stated inputs. Its revenue is 120. The car company's total expenses are 200. GDP is 250, and cars are the only final output in the exercise.

1. Car revenue is **250**, equal to GDP.
2. Steel purchases cost 120, so car-sector wages are 200 − 120 = **80**.
3. Car-sector profit is 250 − 200 = **50**.

Check: total income = 80 + 40 + 80 + 50 = 250. Alternatively, value added = 120 + (250 − 120) = 250. These are the requested values (i), (ii), and (iii): **250, 80, 50**.

2.3 Expenditure components

Equation: $Y = C + I + G + NX$, where $NX = X - IM$.

In this standard identity, I includes inventory investment. When the lecture writes inventory investment separately, I refers to fixed investment only. Use one convention consistently and do not add inventories twice.

Component	What belongs in it	Common error
Consumption C	Household purchases of goods and services, including new consumer cars	Treating every durable good as investment
Fixed investment I	New business plant and equipment; new residential construction	Calling a purchase of shares or existing bonds GDP investment
Government purchases G	Government purchases of goods and services, including employee services	Including cash transfers as direct purchases
Exports X	Domestically produced goods and services sold abroad	Excluding them because the buyer is foreign
Imports IM	Foreign production included in domestic spending	Forgetting to subtract imports already counted in C, I, or G
Inventory investment	Change in stocks of produced but unsold goods	Equating production with sales in every period

Imports are subtracted to remove foreign production from domestic spending. An imported consumer good can raise C and IM by the same amount, leaving domestic GDP unchanged by that imported production. Domestic distribution services may still add domestic value.

Government transfers redistribute purchasing power without directly purchasing newly produced output. In the model they affect disposable income through net taxes T. If a recipient subsequently buys a service, that purchase enters the appropriate expenditure category. The same transfer must not be counted again as G.

Inventories reconcile production and sales

Suppose a firm produces 100 worth of goods but sells 80. Inventory investment is +20, so GDP is 100 rather than 80. If next period it sells the remaining 20 without producing anything new, expenditure on those goods is +20 and inventory investment is −20. Their net contribution to next period's GDP is zero.

Equation: Inventory investment = production − sales, using consistent valuation.

In the course's simplified equilibrium model, unintended inventory changes disappear when Y = Z. Small net inventory investment or net exports in a particular dataset is a motivation for simplifying the model, not proof that inventories or international trade are unimportant. Small NX can coexist with very large exports and imports.

2.4 GDP and welfare

GDP measures production. It does not directly deduct pollution, describe inequality, measure leisure, or fully capture unpaid care and volunteer work. Repairs after damage can raise GDP without making society better off overall.

The tutorials' pollution and happiness examples show why GDP per capita is informative about material resources but incomplete as a welfare measure. Correlation between income and life satisfaction also does not establish causation.

The slides overgeneralise when excluding activities without money. Some services are imputed, notably owner-occupied housing, and non-cash production can count. Unreported production creates measurement problems; tax reporting does not define the production boundary. In exercises, identify current domestic production and apply the stated accounting treatment.

3. Real GDP, chain indexes, and inflation

Sources: L2-3, slides 24-32 and 41-49; T1, pages 15-20.

3.1 Separating quantity from price

Nominal GDP uses each year's prices and quantities. It can increase because the economy produces more, because prices rise, or both. Real GDP aims to measure the change in production while separating price changes.

Equations: Nominal $GDP_t = \Sigma_j p_{jt}q_{jt}$.
Fixed-base real $GDP_t = \Sigma_j p_{jb}q_{jt}$, where b is the base year.

The subscript j identifies a good, and t identifies time. Fixed-base real GDP uses the base year's prices to value every year's quantities. Nominal and real GDP are equal in the base year by construction. Outside the base year, either can be larger, depending on prices relative to the base.

Lecture example: potatoes and wine

Item	Year 0 quantity	Year 0 price	Year 1 quantity	Year 1 price
Potatoes	10	1	15	1
Wine	5	2	5	3

Nominal GDP is 20 in year 0 and 30 in year 1. At year 0 prices, real GDP rises from 20 to 25, giving growth of 25%. At year 1 prices, real GDP rises from 25 to 30, giving growth of 20%.

Why do the growth rates differ? The quantity of potatoes rises while wine production is unchanged. Year 1 prices give wine a larger weight, so the growing component has a smaller relative weight. Real growth in a multi-good economy depends on how heterogeneous quantities are weighted.

中文理解: 实际 GDP
仍然需要用价格作权重。固定价格是为了排除当期涨价的影响，不代表“价格从计算里消失了”。

3.2 Fisher quantity growth and chain linking

A distant fixed base can become unrepresentative as relative prices, product quality, and the range of goods change. The course addresses the choice of weights using adjacent-year prices and the geometric average of two quantity ratios.

For the link from t − 1 to t, calculate:

Previous-price ratio: $R^L_t = (\Sigma_j p_{j,t-1}q_{jt})/(\Sigma_j p_{j,t-1}q_{j,t-1})$.

Current-price ratio: $R^P_t = (\Sigma_j p_{jt}q_{jt})/(\Sigma_j p_{jt}q_{j,t-1})$.

Fisher quantity factor: $F_t = \sqrt{(R^L_t \times R^P_t)}$. Real growth $g_t = F_t - 1$.

Take the geometric average of the **growth factors**, not the growth rates. The names Laspeyres and Paasche are useful labels for the two ratios, but the actual price-quantity combinations are more important to remember.

For potatoes and wine, $F_1 = \sqrt{[(25/20) \times (30/25)]} = \sqrt{1.5} = 1.2247449$. Real growth is **22.4745%**. With year 0 as the reference year, chained real $GDP_1 = 20 \times 1.2247449 =$ **24.4949**.

Converting growth factors into a level series

Set chained real GDP in reference year r equal to that year's nominal GDP. Move forward by multiplying successive Fisher factors, and backward by dividing.

Equations: $Y^{chain}_r = \text{nominal } GDP_r$; $Y^{chain}_t = Y^{chain}_{t-1} F_t$.
Going backward: $Y^{chain}_{t-1} = Y^{chain}_t / F_t$.

For a future year r + k, multiply nominal GDP_r by $F_{r+1} \times F_{r+2} \times \ldots \times F_{r+k}$. This notation avoids the ambiguous subscripts in the slide's chain formula.

Changing the reference year rescales an already constructed chain series. It does not change its adjacent-year growth rates. This differs from changing the fixed price weights in a fixed-base series, which can change measured growth. Chained component levels also need not sum exactly to aggregate chained GDP outside the reference year; they use different changing weights.

中文理解: “固定基年法”是一直用同一套价格作权重；“链式法的参考年”主要决定整条序列的标尺。链式增长率先由相邻两年的价格和数量算出，再串起来。

3.3 Tutorial 1: complete three-year chain calculation

Item	Year 1	Year 2, reference	Year 3
Quantity of Good 1	75	76	77
Quantity of Good 2	25	30	35
Price of Good 1	20	21	22
Price of Good 2	60	59	58
Nominal GDP	3,000	3,366	3,724

First form the price-quantity cross-valuations. Keeping prices fixed across each row prevents mismatched numerator and denominator weights.

Valuation prices	Year 1 quantities	Year 2 quantities	Year 3 quantities
Year 1 prices	3,000	3,320	3,640
Year 2 prices	3,050	3,366	3,682
Year 3 prices	3,100	3,412	3,724

Year 1 to year 2: $R^L_2 = 3320/3000 = 1.1066667$ and $R^P_2 = 3366/3050 = 1.1036066$. Therefore $F_2 =$ **1.1051355528**, so growth is **10.5136%**.

Year 2 to year 3: $R^L_3 = 3682/3366 = 1.0938800$ and $R^P_3 = 3724/3412 = 1.0914420$. Therefore $F_3 =$ **1.0926602929**, so growth is **9.2660%**.

Because year 2 is the reference year, chained real GDP_2 = 3,366. Chained real GDP_1 = 3366/F_2 ≈ **3,045.78**. Chained real GDP_3 = 3366F_3 ≈ **3,677.89**. Keep full precision until the final rounding. The tutorial's displayed year 1 result of 3,045 is a rough value, not the correctly rounded two-decimal answer.

The number 3,050 is year 1 GDP valued at year 2 prices using the fixed-base method. It is not the same object as year 1 chained real GDP in year 2 reference dollars, 3,045.78.

3.4 GDP deflator and inflation

The GDP deflator is an aggregate price index for domestic production:

Equations: P_t = nominal GDP_t/real GDP_t.
Inflation $\pi_t = (P_t - P_{t-1})/P_{t-1} = P_t/P_{t-1} - 1$.

The slides normalise P to 1 in the base or reference year. Many published tables instead use 100. Both conventions produce the same inflation rate because multiplying every index value by 100 cancels in the ratio.

The numerical level of a price index depends on the normalisation. Its ratios and percentage changes are economically meaningful. A value of 1.20 on a base-1 index means a price level 20% above its reference value. An index number alone is not a currency price for the whole economy.

Inflation means $\pi > 0$. Deflation means $\pi < 0$. **Disinflation** means inflation is falling: if inflation declines from 8% to 3%, prices are still rising, but more slowly. A lower inflation rate does not itself mean a lower price level.

Cars: fixed-base real GDP and the deflator

The lecture uses years 2008-2010 with 2009 as base; Tutorial 1 uses the same data labelled 2011-2013 with 2012 as base. The calculations are identical.

Period	Quantity	Price	Nominal GDP	Real GDP at 24,000	Deflator, base = 1
First year	10	20,000	200,000	240,000	0.833333
Base year	12	24,000	288,000	288,000	1.000000
Third year	13	26,000	338,000	312,000	1.083333

Real growth is 20% and then 8.3333%. Inflation is 20% and then 8.3333%. These happen to match numerically in this example but need not do so generally. Nominal GDP growth is 44% and then 17.3611%.

Because there is only one good, the deflator equals that good's price divided by its base-year price. In a multi-good economy, the GDP deflator is an aggregate index, not the relative price between two individual goods.

Exact relationship between nominal and real growth

Since nominal GDP = P × real GDP:

Equation: $1 + g^N = (1 + \pi)(1 + g^R)$.

Thus $g^N = \pi + g^R + \pi g^R$. The familiar rule $g^N \approx \pi + g^R$ is an approximation. For 20% real growth and 20% inflation, nominal growth is 1.2 × 1.2 − 1 = 44%, not 40%.

3.5 Consumer price index

The CPI measures the price of a representative consumer basket. In the course's fixed-basket construction, choose quantities for a basket, price that same basket each year, and divide by its base-year cost.

Equation: $CPI_t = (\Sigma_j p_{jt} q_j^{basket})/(\Sigma_j p_{jb} q_j^{basket})$, optionally multiplied by 100.

Added illustration: a basket costs 200 in the base year and 220 next year. CPI rises from 100 to 110, and inflation is 10%. The basket quantities remain fixed during this comparison.

Question	GDP deflator	CPI in the basic course construction
Which goods?	Domestic final production	Representative household purchases
Imported consumer goods?	Excluded as foreign production	Included if in the consumer basket
Exports?	Included as domestic production	Generally outside the domestic consumer basket
Domestic capital or military equipment?	Included when part of final domestic production	Generally excluded from household consumption
Weights?	Based on the output measure, with quantities changing	Fixed basket quantities in the simple calculation

An imported oil price increase can raise consumer prices substantially without directly raising the domestic-output deflator through the imported oil itself. Indirect effects on domestic production costs can still matter. “Core inflation” commonly removes food and energy categories to help distinguish their volatility from other price movements; it does not mean those costs are irrelevant to households.

3.6 Why inflation matters

If every price and nominal income rose in exactly the same proportion, purchasing power and relative prices would be unchanged in a very simplified thought experiment. Actual inflation is uneven. Fixed nominal pensions or incomes can lose purchasing power, while different goods' prices adjust at different speeds.

Relative-price movements normally convey information about scarcity and demand. Uneven price adjustment during inflation can obscure those signals and affect spending or investment decisions. The lecture's price-rigidity evidence also prepares the fixed-price assumption used later in IS-LM.

4. Employment, unemployment, and participation

Source: L2-3, slides 34-39.

4.1 The definitions fit together

Employment N counts people with jobs. Unemployment U counts people without work who satisfy the relevant job-search and availability criteria. The course simplifies this to people without jobs who are looking for work. The labour force L consists of the employed plus the unemployed.

> **Equations:** L = N + U; unemployment rate u = U/L.
> Participation rate = L/working-age population.

The denominator of the unemployment rate is the labour force, not the entire population. A person without a job is not automatically classified as unemployed. A student, retiree, or discouraged worker may be outside the labour force if they do not meet the search criteria. National statistical definitions determine the precise population and age coverage.

Discouraged workers want work but stop searching because they believe suitable work is unavailable. They leave measured unemployment and the labour force under the course definition. This can lower the unemployment rate without creating a single job.

4.2 Worked interpretation example

Added illustration: a working-age population of 1,000 contains 570 employed people, 30 unemployed people, and 400 outside the labour force. Then L = 600, u = 30/600 = **5%**, and participation = 600/1000 = **60%**.

If 10 unemployed people stop searching, employment stays at 570, unemployment falls to 20, and the labour force falls to 590. The unemployment rate becomes 20/590 ≈ **3.39%**, and participation falls to **59%**. The headline unemployment rate improves while employment does not change.

If those 10 people instead find jobs, employment rises to 580, unemployment falls to 20, and L remains 600. The unemployment rate becomes **3.33%**, with participation still **60%**. Similar unemployment-rate movements can describe quite different labour-market developments.

> 中文理解: 失业率下降，可能是找到了工作，也可能是放弃找工作。要同时看就业人数和劳动参与率，不能只看一个百分比。

4.3 Why the measure matters

High unemployment often accompanies weak demand and unused productive resources. Job losses can also have lasting earnings effects, as the lecture's layoff study illustrates. GDP and unemployment therefore provide complementary information: aggregate production and the use of labour.

The supplied lectures do not yet specify a production function or a quantitative equation linking Y to u. IS-LM predicts output effects under its assumptions; it does not by itself calculate an unemployment rate or the natural rate of unemployment. Those belong to the later labour-market material announced at the end of Lecture 8-9.

5. The goods market and the Keynesian cross

Sources: L4-5, slides 14-28; L6-7, slides 2-4; T2, pages 9-18.

5.1 Consumption responds to disposable income

The course assumes a linear aggregate consumption function:

> **Equation:** $C = c_0 + c_1 Y^D = c_0 + c_1(Y - T)$, with $0 < c_1 < 1$.

c_0, autonomous consumption, is consumption when disposable income is zero in this linear function. It also captures shifts associated with wealth, confidence, expected future income, access to credit, and other influences held outside the model. Positive consumption at zero current income can be financed through assets, borrowing, or transfers already reflected in net taxes.

c_1, the marginal propensity to consume (MPC), measures the increase in consumption from one extra unit of disposable income. If $c_1 = 0.6$, an extra 100 of disposable income raises consumption by 60 and saving by 40, holding other influences fixed.

MPC is a marginal response, not the average fraction of income consumed. The average propensity to consume is $C/Y^D = c_0/Y^D + c_1$ when Y^D is positive. It can exceed MPC, and consumption can exceed current disposable income.

The lecture discusses heterogeneity: households facing tighter resources often consume a larger fraction of an extra unit of income. The model's aggregate c_1 simplifies these different responses. Use the parameter specified in the question rather than assuming one universal empirical MPC.

Movement versus shift

On a graph with Y^D horizontally and C vertically, changing Y^D moves along the consumption line. Changing c_0 shifts the line vertically. Changing c_1 changes its slope.

On a graph with Y horizontally, a rise in T lowers the intercept $c_0 - c_1 T$. On the C-versus-Y^D graph, that same tax change reduces Y^D and causes a movement along the line. A correct answer must say which axes are being used.

> 中文理解: 加税到底是“沿线移动”还是“曲线移动”，取决于横轴。如果横轴是可支配收入，税收通过横轴变化起作用；如果横轴是总收入，固定总收入时消费更低，因此曲线下移。

5.2 Planned demand and equilibrium

In the first closed-economy model, investment is fixed at $\bar{I}$ and net exports are zero:

> **Equation:** $Z = C + \bar{I} + G = c_0 - c_1 T + \bar{I} + G + c_1 Y$.

Define autonomous expenditure $A = c_0 - c_1 T + \bar{I} + G$. Then $Z = A + c_1 Y$. “Autonomous” means independent of current Y in this particular equation, not permanently unchanging.

Equilibrium requires $Y = Z$. Substitution gives:

> **Derivation:** $Y = A + c_1 Y$
> $(1 - c_1)Y = A$
> $Y^* = A/(1 - c_1) = (c_0 + \bar{I} + G - c_1 T)/(1 - c_1)$.

The asterisk marks the equilibrium value. Consumption is endogenous because it responds to equilibrium income. In this model c_0, c_1, $\bar{I}$, G, and T are given.

All terms must use consistent real units. In the goods-market model, Y is real output and real income. If nominal income is PY, dividing by P converts it to real purchasing power. Setting P = 1 is a convenient normalisation, not a reason to mix nominal income with real expenditure.

5.3 Reading the Keynesian-cross diagram

Put income/output Y on the horizontal axis and demand Z and production on the vertical axis. The 45-degree line has height Y and slope 1. The ZZ line has intercept A and slope c_1. Their intersection is equilibrium.

If Y is below equilibrium, ZZ lies above the 45-degree line: planned demand exceeds current production. Inventories are unexpectedly depleted, which encourages firms to increase output. If Y is above equilibrium, output exceeds planned demand, inventories accumulate unexpectedly, and firms reduce production.

The condition $c_1 < 1$ ensures that planned demand rises by less than output when Y increases. It also makes the simple multiplier series converge. A change in autonomous expenditure shifts ZZ in parallel if c_1 is unchanged. The vertical shift in ZZ is the initial spending change; the horizontal change in equilibrium Y is the total multiplied response.

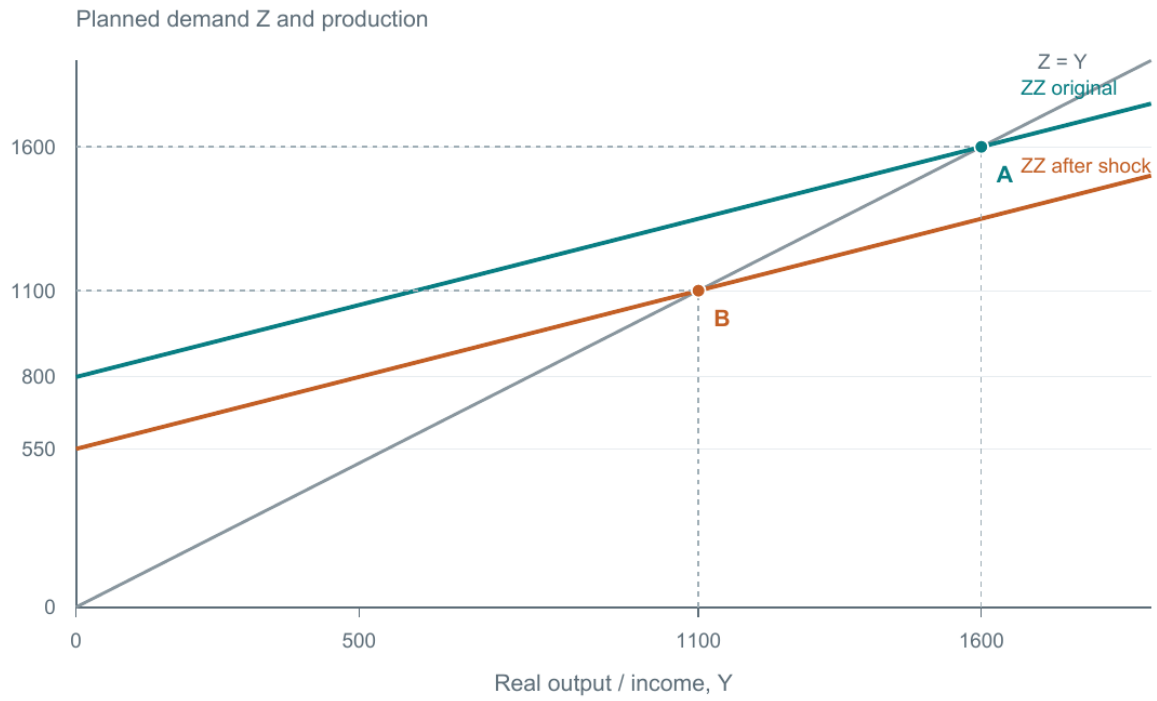


Figure 1. An illustrative fall in autonomous expenditure shifts ZZ down. The decline in equilibrium output exceeds the initial spending reduction because consumption also falls.

5.4 Why the multiplier exists

Suppose c_0 increases by 100. Buyers initially demand 100 more output. Firms expand production, generating 100 more income. Households spend $c_1 \times 100$ of the extra income, leading to another round of production and income, followed by $c_1^2 \times 100$, and so on.

> **Equation:** $\Delta Y = [1 + c_1 + c_1^2 + ...]\Delta A = \Delta A/(1 - c_1)$.

If $c_1 = 0.5$, the rounds are 100 + 50 + 25 + 12.5 + ... = 200. The multiplier is 2. It is larger when more of each extra income unit is spent, because less leaks into saving at each round.

The same reasoning applies to a fall in autonomous expenditure: the initial loss of sales reduces income, which reduces consumption and sales further. The multiplier amplifies both expansions and contractions.

Do not confuse this spending feedback with the velocity of money. The multiplier follows from income-induced expenditure. Velocity measures nominal expenditure relative to the money stock. They are distinct concepts.

5.5 Tutorial 2: full equilibrium solution

The exercise gives $C = 480 + 0.5Y^D$, $\bar{I} = 105$, $T = 70$, and $G = 250$.

Step 1: express demand as a function of Y.

> $Z = 480 + 0.5(Y - 70) + 105 + 250 = 800 + 0.5Y$.

Step 2: impose equilibrium. $Y = 800 + 0.5Y$, so $0.5Y = 800$ and **$Y^* = 1{,}600$**.

Step 3: calculate disposable income. $Y^D = 1600 - 70 =$ **1,530**.

Step 4: calculate consumption. $C = 480 + 0.5 \times 1530 =$ **1,245**.

Step 5: check goods-market equilibrium. $C + I + G = 1245 + 105 + 250 =$ **1,600**, matching output.

An additional saving check connects to the next chapter. Private saving is $1530 - 1245 =$ **285**. Public saving is $70 - 250 =$ **−180**. National saving is $285 - 180 =$ **105**, equal to investment.

The answer 800 is autonomous expenditure, not equilibrium output. The answer 1,530 is disposable income, not output. Keeping these three objects separate prevents a common mistake.

5.6 Tutorial 2: the pandemic demand shock

The tutorial asks what happens when autonomous consumption falls by 200 and investment falls by 50. Initially, at any given income, planned expenditure falls by 250.

> **Equation:** $\Delta A = \Delta c_0 + \Delta\bar{I} = -250$; $\Delta Y = -250/(1 - c_1)$.

Using $c_1 = 0.5$ from the preceding exercise gives **$\Delta Y = -500$**. If we also retain that exercise's starting values, new output is **1,100**. New $c_0 = 280$, investment = 55, disposable income = 1030, and consumption = $280 + 0.5 \times 1030 =$ **795**. Check: $795 + 55 + 250 = 1100$.

The direct consumption loss is 200. The further consumption loss from lower income is $0.5 \times 500 = 250$. Total consumption therefore falls by 450, while investment falls by 50, together explaining the 500 output decline.

The slide presenting the shock does not restate c_1. The general answer is $-250/(1 - c_1)$; the numerical result −500 assumes the earlier value continues to apply.

The model captures weaker confidence, postponed purchases, and reduced investment. A pandemic can also disrupt supply, force business closures, change sectoral demand, and alter prices. A demand-only fixed-price model cannot explain all of those developments. This limitation is the appropriate response to the tutorial's question about how well the model describes events.

6. Saving, fiscal policy, and stabilisers

Source: L4-5, slides 30-55.

6.1 Private, public, and national saving

Private saving is disposable income not consumed:

> **Equations:** $S = Y - T - C = -c_0 + (1 - c_1)(Y - T)$.
> Public saving $= T - G$. National saving $= S + T - G = Y - C - G$.

The **marginal propensity to save** is $1 - c_1$. Public saving is positive for a surplus and negative for a deficit in this simple model. With T defined as net taxes, transfers have already been included in the tax-and-transfer balance.

Rearranging goods-market equilibrium gives the investment-saving relation:

> **Derivation:** $Y = C + I + G$
> $I = Y - C - G = (Y - T - C) + (T - G)$
> $I = S + (T - G)$.

This is the same goods-market condition viewed through saving. It does not say private saving alone always equals investment. That would require $T = G$. It also applies here to a closed economy; international lending and borrowing change the relationship in an open economy.

Actual saving and actual investment, including unintended inventory changes, satisfy the accounting identity. The equilibrium condition says planned saving and planned investment are compatible without unintended inventories. The equality alone does not establish which variable drives the adjustment.

6.2 The paradox of thrift

Suppose all households try to save more at any given income by reducing c_0. With $\bar{I}$, G, and T fixed, demand and equilibrium income fall. Because investment and public saving are fixed, equilibrium private saving must remain:

> **Equation:** $S = \bar{I} - (T - G)$, a fixed amount in this model.

For a reduction in c_0 of 1, $\Delta c_0 = -1$ and $\Delta Y = -1/(1 - c_1)$. The change in saving is:

> $\Delta S = -\Delta c_0 + (1 - c_1)\Delta Y = 1 - 1 = 0$.

The initial desire to save more is offset by the loss of income. Each household's intention is understandable; the aggregate result reflects that everyone's lower spending reduces everyone else's sales and income.

> **中文理解:** 大家都想在“原来的收入”下多存钱，但减少消费后，总收入也下降了。最后收入下降抵消了想多储蓄的效果。这并不代表个人储蓄没有意义。

The lecture also asks about reducing c_1. Under fixed investment and fiscal variables, equilibrium private saving is still fixed by $S = \bar{I} - T + G$. For positive disposable income, a lower c_1 reduces consumption demand and equilibrium output. Differentiating the equilibrium relation gives $dY/dc_1 = (Y - T)/(1 - c_1)$, which is positive when $Y - T > 0$.

This is a short-run result under specific assumptions. If investment responds to income and interest rates, public saving changes, or productive capacity adjusts over time, the final saving outcome can change. Do not carry “saving never rises” into every model.

6.3 What governments spend on

The lecture distinguishes public goods such as infrastructure and safety, social programmes such as education and retirement support, and measures intended to stabilise economic activity. Its historical charts show the expansion of the welfare state and changes in spending composition. Those charts provide context for the importance of fiscal policy; their percentages are historical examples rather than fixed parameters of the model.

Mandatory or entitlement spending follows eligibility and benefit rules, so spending can change as the number of recipients changes. **Discretionary spending** is set through budget decisions. The distinction does not mean mandatory programmes can never be changed: legislators can change their rules. It also does not mean every mandatory programme responds strongly to recessions.

Government budget outlays are broader than G in the GDP equation. Transfers and interest payments are not direct purchases of current goods and services. Keep the institutional budget discussion separate from the simplified expenditure variable G.

6.4 Spending, tax, and balanced-budget multipliers

With c_1 unchanged, changes in the Keynesian-cross equilibrium satisfy:

> **Master equation:** $\Delta Y = [\Delta c_0 + \Delta \bar{I} + \Delta G - c_1 \Delta T]/(1 - c_1)$.

Change, holding other exogenous variables fixed	Multiplier	Initial effect on demand
Higher government purchases	$\Delta Y/\Delta G = 1/(1 - c_1)$	One unit for each unit of G
Higher taxes	$\Delta Y/\Delta T = -c_1/(1 - c_1)$	Consumption falls by c_1 per tax unit
Tax cut of one unit	Output rises by $c_1/(1 - c_1)$	Households spend only the MPC fraction initially
Equal increases in G and T	$\Delta Y/\Delta G = 1$, when $\Delta G = \Delta T$	Initial net demand rises by $1 - c_1$ per unit
Higher c_0 or fixed investment	$1/(1 - c_1)$	Direct addition to autonomous demand

Why a tax cut has a smaller effect than equal purchases

An extra unit of G directly buys one unit of output. A one-unit tax cut raises disposable income by one, but only c_1 is initially spent; the rest is saved. Both initial effects then pass through the same consumption feedback. The spending multiplier therefore exceeds the absolute tax multiplier by exactly 1 in this model.

Why the balanced-budget multiplier is positive

If $\Delta G = \Delta T = x$, substitute into the master equation:

$$\Delta Y = (x - c_1 x)/(1 - c_1) = x.$$

Government purchases add x directly, whereas higher taxes initially reduce consumption by $c_1 x$. The net initial demand increase is $(1 - c_1)x$. Multiplication by $1/(1 - c_1)$ gives x.

The government's initial budget need not be balanced. Equal increases in purchases and taxes leave its existing balance unchanged. “Balanced-budget multiplier” refers to the financing of the change.

In this fixed-investment model, $\Delta Y = \Delta T$ implies disposable income is unchanged after the adjustment. Consequently $\Delta C = 0$, $\Delta S = 0$, and public saving is unchanged. The increase in output equals the increase in G.

Source correction: L4-5, slides 46-47, contain a downward tax arrow in the first feedback line. The stated experiment requires **G rises and T rises by the same amount**.

Numerical comparison using the Tutorial 2 baseline

Start from $c_1 = 0.5$ and $Y = 1600$.

Policy	Change in Y	New Y	Change in C	Change in T − G
G rises by 100	+200	1,800	+100	−100
T falls by 100	+100	1,700	+100	−100
G and T both rise by 100	+100	1,700	0	0

These are separate experiments. Do not accumulate them unless a question explicitly asks for a sequence.

6.5 Automatic stabilisers

Automatic stabilisers are existing rules that cushion fluctuations without requiring a new discretionary policy decision each time. When income falls, tax collections tend to fall and eligibility for some transfers rises. Disposable income therefore falls by less than pre-tax income, limiting the decline in consumption.

Unemployment insurance supports the income of eligible people who lose their jobs. Progressive taxes collect more as incomes rise and less as incomes fall. Both can dampen the feedback between output and consumption. An automatic increase in benefits can widen a deficit even if no new stimulus legislation has passed.

Algebraic extension of the lecture's intuition

Let net taxes be $T = T_0 + tY$, with $0 < t < 1$ and investment still fixed. Here t is the marginal response of net taxes to income. This linear rule illustrates the stabilising mechanism; a constant t need not be a progressive tax system by itself.

Then disposable income is $(1 - t)Y - T_0$, and:

Equations: $Z = c_0 - c_1 T_0 + \bar{I} + G + c_1(1 - t)Y$.
$Y^* = (c_0 - c_1 T_0 + \bar{I} + G)/[1 - c_1(1 - t)]$.

The spending multiplier becomes $1/[1 - c_1(1 - t)]$. This is smaller than $1/(1 - c_1)$, because some additional income automatically becomes net taxes instead of disposable income. With $c_1 = 0.8$ and $t = 0.25$, the multiplier falls from 5 to 2.5.

The new autonomous-tax multiplier is $-c_1/[1 - c_1(1 - t)]$. The earlier balanced-budget result of exactly 1 cannot be applied mechanically to changes in T_0: total taxes now also change when output changes.

6.6 Policy lags and model limits

Discretionary fiscal measures can take time to recognise, approve, and implement. Automatic stabilisers respond through existing rules. Monetary decisions may be taken more quickly, although their effects on spending also take time.

The basic multiplier holds investment, interest rates, prices, and external trade fixed. Real outcomes can differ because these variables respond, because households have different MPCs, because policy is financed differently, or because capacity is constrained. The lecture cites a historical review suggesting a central range of roughly 0.8-1.5 for certain temporary, deficit-financed government-purchase multipliers, with wider uncertainty. This is evidence about specified settings, not a universal constant.

At a zero interest-rate bound, conventional monetary easing may have limited additional scope, making fiscal demand support especially relevant within this framework. IS-LM now adds the financial feedback that the first multiplier model leaves out.

7. Money, bonds, and monetary policy

Sources: L6-7, slides 7-37; T3, pages 3-15.

7.1 The portfolio choice

Money is a liquid asset used for transactions. The course describes M1 as currency plus readily usable deposits, and M2 as a broader aggregate including some less liquid assets. These are stylised descriptions; exact aggregate definitions depend on the statistical system. Their role in the model is to distinguish readily spendable balances from broader wealth.

Given financial wealth, households choose how much to hold as money and how much as bonds. Money provides liquidity but pays zero nominal interest in the model. Bonds pay interest but are not directly usable for ordinary purchases. Holding money therefore has an opportunity cost: the interest forgone on bonds.

Model asset	Direct transaction use	Nominal return	Reason to hold it
Money	Yes	0	Liquidity
One-year risk-free bond	Must first be converted to money	i	Interest income

This completes the comparison table in Tutorial 3. Money demand is a desired **stock of liquid assets**, not how much income a person wants to earn or how much consumption they want to buy.

7.2 Bond prices and yields

For a one-year zero-coupon bond, the holder pays P^B today and receives the face value F in one year. There are no coupon payments during the year.

Equations: $i = (F - P^B)/P^B = F/P^B - 1$.
$P^B = F/(1 + i)$.

With a fixed future payoff, a higher purchase price gives a lower return. The bond-price-interest-rate relationship is therefore inverse. The denominator in the return formula is the amount invested today, not the face value received later.

Tutorial 3: a bond paying 100

Price today	Calculation	One-year yield
98	(100 − 98)/98	2.0408%
99	(100 − 99)/99	1.0101%
97	(100 − 97)/97	3.0928%

The lecture rounds the price-99 yield to 1%; its exact value is approximately 1.0101%. If a question specifies a 5% yield, the price is 100/1.05 ≈ 95.2381. Enter 0.05, not 5, into the formula.

The lecture also shows a coupon bond with face value 5,000 and annual coupon rate 8%, paid semiannually. It pays 5,000 × 0.08/2 = **200 every half-year**, plus the principal at maturity. A coupon rate is a payment rule based on face value; it need not equal the market yield when the price changes. The one-year zero-coupon formula is the formula used in the core model.

7.3 Money demand

The money-demand equation is:

Equation: $M^d = PY \times L(i)$, with $L(i) > 0$ and $L'(i) < 0$ in the relevant range.

Nominal income PY proxies for the value of transactions. More real activity raises transactions; higher prices make a given quantity of transactions require more nominal spending. Both increase desired nominal money holdings at a given interest rate.

The function L(i) measures desired money holdings relative to nominal income. A higher interest rate makes holding bonds more attractive and holding non-interest-bearing money more costly, so L(i) falls.

With i vertically and M horizontally, money demand slopes downward. An interest-rate change moves along the curve when nominal income is fixed. Higher nominal income shifts the curve to the right. A change in payment technology or liquidity preference can also shift the function itself.

中文理解: 货币需求不是“想消费多少”，而是“为了交易和流动性，想把多少财富以货币形式拿在手里”。利率越高，持有不生息货币的机会成本越高。

7.4 Money supply and equilibrium

The central bank sets $M^s = M$ in the simple model. Money supply is vertical in the (M, i) diagram because the supplied quantity is fixed rather than chosen as a function of i within that experiment.

Equilibrium: $M = M^d = PY\,L(i)$.

If the central bank raises M while PY is fixed, the equilibrium interest rate falls. At the old rate, people collectively hold more money than they wish to. Attempts to buy bonds raise bond prices and reduce yields until people willingly hold the larger money stock.

An individual can exchange money for bonds, but that transaction transfers money to another person. It does not eliminate aggregate money. The adjustment in bond prices and interest rates changes the public's desired aggregate holdings.

If PY increases while M is fixed, money demand shifts right and i rises. People seek more liquid balances and sell or demand fewer bonds. Bond prices fall, raising yields. The higher opportunity cost of money reduces desired holdings enough to restore equilibrium.

Tutorial 3, page 6: numerical equilibrium

The image-based exercise gives $M^d = \$Y(0.2 - i)$, nominal income $\$Y = 2000$, and $M^s = 300$. The stated monetary units are trillions of US dollars.

$300 = 2000(0.2 - i) = 400 - 2000i$.
$2000i = 100$, so **$i = 0.05 = 5\%$**.

For the graph, invert demand: $i = 0.2 - M^d/2000$. Its vertical intercept is 20% and its horizontal intercept is 400. Draw supply vertically at M = 300 and mark equilibrium at (300, 5%). The equation is economically relevant over the range where desired money is non-negative.

Added comparative-static checks: with nominal income still 2000, raising money supply to 320 gives i = 4%. With M still 300, raising nominal income to 2400 gives i = 7.5%. These isolate the two distinct shifts.

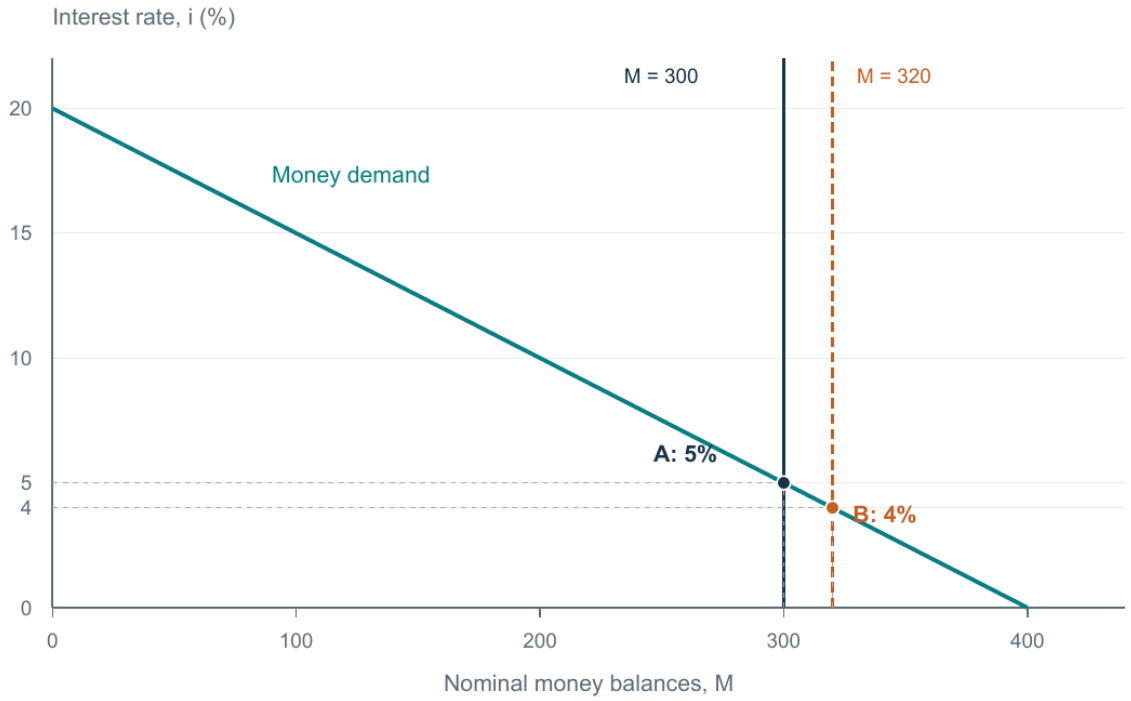


Figure 2. Tutorial 3's original equilibrium is M = 300 and i = 5%. An illustrative increase in M to 320 lowers i to 4% at unchanged nominal income.

7.5 Open market operations: the same change in two markets

An open market operation is a central-bank purchase or sale of bonds in the market. In the course model, the central bank pays for a purchase by supplying money and absorbs money when it sells bonds.

Operation	Money in circulation	Bond-market pressure	Bond price	Interest rate
Central bank buys bonds	Increases	Additional demand for bonds	Rises	Falls
Central bank sells bonds	Decreases	More bonds offered / less central-bank demand	Falls	Rises

Expansion: The central bank buys bonds; sellers receive money; M rises. The money-supply line shifts right. Bond prices rise and interest rates fall.

Contraction: The central bank sells bonds; buyers pay money to it; M falls. The money-supply line shifts left. Bond prices fall and interest rates rise.

These fill Tutorial 3's blanks: to lower rates the central bank **increases** money supply; bond sales **remove or reduce** money in circulation; supply **decreases** and shifts **left**.

中文记忆:
央行买债，把钱付给市场；央行卖债，把钱收回。先确认交易主体是“央行”，再判断货币供给方向。

Why the tutorial's bond-demand curve slopes upward

Tutorial 3 draws i on the vertical axis. A higher yield corresponds to a lower bond price and makes bonds more attractive, so desired bond holdings rise. The bond-demand curve is therefore upward-sloping in this diagram. With bond price vertically, the usual demand curve slopes downward.

An increase in nominal income shifts money demand right and, at given financial wealth, shifts private bond demand left. In the tutorial's interest-rate diagram, a leftward shift of the upward-sloping bond-demand curve raises the equilibrium yield at fixed bond supply.

A central-bank purchase can be represented as additional total demand for bonds, shifting that demand right and reducing i. Alternatively, if the graph includes only private holdings, the central bank's purchase reduces the supply of bonds left for the public. These are alternative accounting conventions for the same transaction; do not apply both shifts to one graph and double-count it.

The tutorial's historical rate chart suggests expansionary policy during the sharp 2008-2009 rate reductions and contractionary policy during the 2016-2018 increases. Within the simple model these correspond to bond purchases and bond sales respectively. The chart shows the policy-rate stance, not a transaction-by-transaction record of the central bank's actual operating tools.

7.6 Velocity and the quantity equation

Velocity measures nominal expenditure relative to the money stock:

> **Equations:** V = PY/M; MV = PY. At money-market equilibrium, V = 1/L(i).

If nominal GDP is 1000 and the money stock is 200, velocity is 5 per period. A unit of money can support multiple transactions as it moves between holders. It is therefore unnecessary for the money stock to equal a year's total spending.

Payment technologies can reduce the quantity of balances needed for a given level of transactions. This is the intuition behind the lecture's discussion of shifts in M/PY over time. It does not establish that velocity must always rise in every period.

MV = PY is an identity when V is defined as PY/M. A theory predicting that more money only raises prices requires additional assumptions about V and real output. In short-run IS-LM, P is fixed and Y can respond; do not import a fixed-output conclusion from a different model.

7.7 The liquidity trap and zero lower bound

In the stylised model, money pays zero nominal interest and is more liquid than bonds. If a comparable bond offered a negative nominal return, costlessly held money would dominate it. This gives the model a zero lower bound on i.

At i = 0, once transaction needs are met, holders are willing to absorb extra money instead of bonds without a further fall in the interest rate. Money demand becomes horizontal at zero in the usual diagram, and a further expansion of money supply can leave i unchanged. This is the liquidity trap.

The limitation concerns the conventional channel “more money lowers i, which raises investment.” It does not logically imply that all forms of policy or every change in financial conditions are ineffective. The literal zero bound depends on the model's assumptions about money, returns, and storage costs.

The lecture uses Japan's low-yield history and the US low-rate episodes as illustrations. A chart of a long-term bond yield and a chart of a short-term policy rate are not interchangeable measures, even though both can illustrate low nominal interest rates.

> 中文理解: 流动性陷阱中，新增货币可能只是被持有，并没有继续压低模型中的利率。因此，“增加货币”到“增加投资”的这条传导链会受限。

8. Building the IS-LM model

Source: L8-9, slides 1-24.

8.1 What changes from the Keynesian cross

Investment is now a function of output and the interest rate:

> **Equation:** $I = I(Y, i)$, with $I_Y > 0$ and $I_i < 0$.

Higher output means higher sales and greater desired productive capacity, encouraging investment. A higher interest rate raises the cost of financing investment and makes fewer projects attractive. The lecture's historical chart illustrates that investment and output tend to move together and that investment is relatively volatile.

Investment reacts to two variables. A policy that raises output and raises interest rates creates opposing investment effects. A policy that raises output and lowers interest rates makes both effects expansionary. This distinction explains many of the policy results later.

Planned demand becomes $Z = C(Y - T) + I(Y, i) + G$. Holding i fixed, its slope with respect to Y is MPC + MPI, where MPI is the marginal propensity to invest out of output, I_Y. The lecture assumes $\mathbf{c_1 + I_Y < 1}$, so demand still rises less than one-for-one with output and the usual stable crossing remains.

The model uses the nominal interest rate i in this investment relation. A more detailed treatment distinguishes real borrowing costs and inflation expectations; those extensions are not developed in the supplied lectures.

8.2 The IS curve: goods-market equilibria

The IS relation is:

> **Equation:** $Y = C(Y - T) + I(Y, i) + G$.

The IS curve contains combinations of Y and i that satisfy this equation for given fiscal policy and other spending determinants. To derive its direction:

1. Start from a goods-market equilibrium at an initial interest rate.
2. Raise i while holding the other exogenous variables fixed.
3. Investment falls at every given Y, shifting ZZ down.
4. Lower demand reduces output and income, causing additional consumption and investment reductions.
5. The new goods-market equilibrium has lower Y.

Thus the IS curve slopes downward in a diagram with Y horizontally and i vertically. A change in i creates a **movement along IS**. A tax increase, a change in government purchases, or a change in confidence changes equilibrium output at every i and therefore **shifts IS**.

Change	IS shift	Reason at a given interest rate
G rises	Right	Higher direct demand
T falls	Right	Higher disposable income and consumption
c_0 rises	Right	Higher consumption at each income
Autonomous investment rises	Right	Higher investment at each Y and i
Reverse of these changes	Left	Lower autonomous demand

The IS curve is an equilibrium locus, not an ordinary demand curve for one commodity. Its slope reflects the feedback of income on spending as well as the response of investment to i.

The investment-saving derivation

Goods equilibrium can also be written $I(Y, i) = S(Y - T) + T - G$. At fixed i, the investment schedule rises with Y at slope I_Y. National saving rises with Y at slope $1 - c_1$, when T and G are fixed.

The condition $c_1 + I_Y < 1$ is equivalent to $\mathbf{1 - c_1 > I_Y}$. The saving schedule is therefore steeper than the investment schedule. A fall in i shifts investment upward at every Y. Their new intersection occurs at higher Y, deriving the same downward-sloping IS relationship.

8.3 A linear IS equation makes the slope visible

For an explanatory linear specification, let:

$C = c_0 + c_1(Y - T)$; $I = I_0 + aY - bi$, with $a \geq 0$, $b > 0$, and $c_1 + a < 1$.

Define $A = c_0 - c_1T + I_0 + G$ and $\alpha = 1 - c_1 - a > 0$. Goods-market equilibrium becomes:

IS: $\alpha Y = A - bi$, or $i = A/b - (\alpha/b)Y$.

Its slope in the (Y, i) diagram is $-\alpha/b$. Stronger interest sensitivity b makes the IS curve flatter: a small rate change has a larger output effect. A stronger income-spending feedback also increases the output response to an initial demand change.

At a given interest rate, the multiplier on A is $\mathbf{1/(1 - c_1 - a)}$. This is greater than $1/(1 - c_1)$ when $a > 0$, because additional output now induces both consumption and investment. The ultimate IS-LM multiplier will also include the offsetting interest-rate response.

8.4 The LM curve: money-market equilibria

Divide the nominal money-market condition by the price level:

Equation: $M/P = YL(i)$.

Real money supply M/P is the purchasing power of the money stock. The LM curve contains combinations of Y and i that make desired real money balances equal the given real money supply.

At a higher Y, the public wants more money for transactions at every i. With M/P fixed, i must rise to make bonds more attractive and reduce desired money holdings. Therefore LM slopes upward.

In the underlying money-market diagram, increasing Y shifts the money-demand curve right. In the IS-LM diagram, the same change corresponds to moving along the existing LM curve. These statements are consistent because the two diagrams have different axes.

> 中文理解: “Y 上升使货币需求曲线右移”不等于“LM 右移”。LM 本来就记录了在同一个 M/P 下，不同 Y 所对应的均衡利率。

If M rises with P fixed, real supply increases. At every Y, a lower interest rate is needed to induce the public to hold the larger stock. LM shifts down/right. A reduction in M/P shifts LM up/left. A higher liquidity preference at each i also shifts LM up, while a fall in that preference shifts LM down.

A rise in P would reduce M/P at fixed nominal M, shifting LM up. In the supplied short-run IS-LM exercises, however, P is held fixed unless a question explicitly changes that assumption.

Optional derivative check of the LM slope

Along a fixed-M/P LM curve, differentiating M/P = YL(i) gives 0 = L(i)dY + YL′(i)di. Therefore:

> **Equation:** di/dY = −L(i)/[YL′(i)] > 0.

The numerator is positive and L′ is negative. This recovers the upward slope directly from the money-demand assumptions.

8.5 Why this course's LM differs from the textbook

Policy assumption	What is fixed during the experiment?	LM in the basic diagram	Fiscal expansion's interest effect
Course's standard model	M/P	Upward-sloping	i rises as Y raises money demand
Fixed policy-rate target	i at its target	Horizontal	i stays fixed if the target is maintained

If the central bank fixes i, it must supply the money demanded at that rate. As Y rises, the bank increases M/P to maintain the target. Money supply is then endogenous to the target rather than fixed throughout the fiscal experiment.

A horizontal LM due to an interest-rate target should also be distinguished from a horizontal segment caused by a liquidity trap. In the first case the central bank chooses to maintain a rate. In the second, the model's lower bound limits further rate reductions.

For course questions, use the upward-sloping LM unless a question changes the monetary assumption. Mixing an upward-sloping graph with a fixed-interest-rate multiplier gives inconsistent answers.

8.6 Joint equilibrium

At the intersection of IS and LM, one pair (Y*, i*) satisfies both goods and money equilibrium. A point on IS alone need not clear the money market; a point on LM alone need not clear the goods market.

The useful sequence for any shock is: identify the exogenous change; decide which curve shifts; locate the new intersection; then calculate consumption, investment, and saving from their definitions. A direct effect holding another variable fixed is an intermediate thought experiment, not necessarily the final equilibrium.

9. Policy effects, feedback, and policy mixes

Source: L8-9, slides 26-50. The linear numerical examples in section 9.6 are additional practice.

9.1 Fiscal contraction through higher taxes

Take a rise in T with G and M/P unchanged. At any initial income and interest rate, disposable income and consumption fall. IS shifts left. Because output falls, transactions and money demand fall. At the unchanged money stock, bond demand increases and the interest rate falls. The final equilibrium has **lower Y and lower i**.

The lower interest rate stimulates investment relative to what it would be at the original rate. It partly cushions the contraction. On the lecture's diagram, point D on the new IS at the old interest rate shows the hypothetical goods-market result if i did not adjust. The actual equilibrium A′ lies where the new IS intersects the unchanged LM.

Variable-by-variable analysis is more precise than saying “everything falls”:

Variable	Effect of T rising, other policy variables fixed	Explanation
Y	Falls	IS shifts left
i	Falls	Lower Y reduces money demand
Y − T	Falls	Output falls and taxes rise
C	Falls	Disposable income falls
I	Ambiguous in the general model	Lower sales reduce I; lower i raises I
G	Unchanged	Given in the experiment
Public saving T − G	Rises	T rises and G is unchanged
Private saving S	Falls	$S = -c_0 + (1 - c_1)(Y - T)$
National saving	Ambiguous	It equals I in this closed economy

Source correction: L8-9, slide 36, says public saving is unchanged. It should **increase**. The definition T − G and the deck's speaker notes both support this correction.

Fiscal consolidation reduces the deficit through higher taxes, lower government purchases, or both. The lecture's debt and austerity examples illustrate a tension: a government may want to improve its fiscal balance while also avoiding a short-run fall in output. Debt sustainability cannot be inferred from IS-LM alone, and a deficit does not mechanically guarantee a rising debt-to-GDP ratio because GDP also changes.

9.2 Fiscal expansion and crowding out

A rise in G or fall in T shifts IS right. Output rises, money demand rises, and i rises along the unchanged LM. The interest-rate increase dampens investment relative to the result at a fixed interest rate. This interest-rate channel is called **crowding out**.

It does not prove that total investment must fall. Higher Y increases desired investment through the sales channel, while higher i lowers it through the cost channel. With I = I(Y, i), the net result is ambiguous unless additional information determines which effect dominates.

If investment depends only on i, fiscal expansion lowers investment in the standard upward-LM model. If the sales response is sufficiently strong, investment can rise despite the higher interest rate. The term **crowding in** is used for a rise in investment accompanying the fiscal expansion.

> 中文理解: “存在挤出机制”是指利率上升压低投资的那部分作用；“投资总量最终下降”还要比较收入上升带来的刺激作用。两者不能直接画等号。

The fiscal multiplier in IS-LM is smaller than the multiplier obtained from the **same goods-market model with i held fixed**. If comparing with the earlier fixed-investment Keynesian cross, there are two changes: induced investment can amplify the multiplier, while rising i reduces it. The net comparison with $1/(1 - c_1)$ is not universally determined when both assumptions change.

9.3 Monetary expansion

At fixed P, an increase in M raises real money balances and shifts LM down. At the initial Y, money-market equilibrium would require a lower i. The lower rate stimulates investment and goods demand, raising output. The increase in Y then raises transactions and money demand, partly reversing the initial interest-rate drop.

The final equilibrium has **higher Y and lower i**. In the lecture, D represents the lower rate at the original Y after the money-supply change. At D the goods market is not yet in equilibrium. As Y rises, money demand increases and i rises somewhat from that intermediate value. The final interest rate remains below its original value.

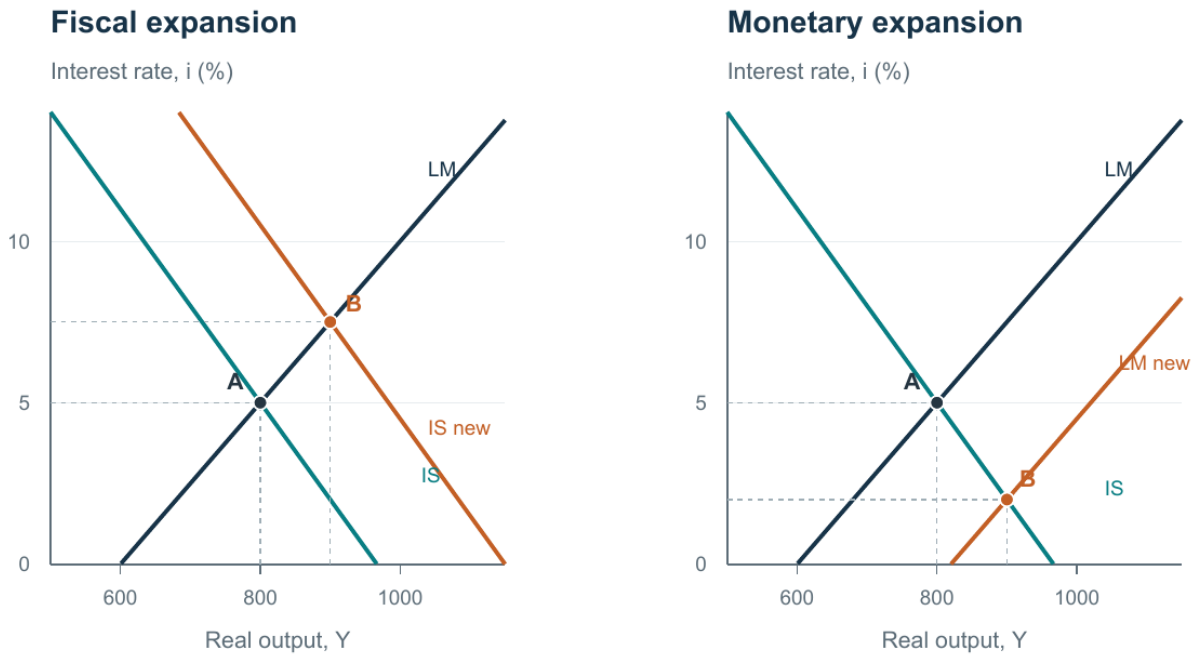


Figure 3. With fixed prices, a fiscal expansion shifts IS right and raises both Y and i. A monetary expansion shifts LM down and raises Y while lowering i. The plotted curves use the illustrative model in section 9.6.

Variable	Effect of M rising, with P, G, and T fixed	Reason
Y	Rises	Lower i stimulates spending
i	Falls	Greater real money supply
C	Rises	Y − T rises
I	Rises	Higher sales and lower i both raise investment
Public saving	Unchanged	T and G unchanged
Private saving	Rises	A fraction $1 - c_1$ of extra disposable income is saved

A monetary contraction reverses these directions away from the lower bound. Monetary expansion is more reliably investment-supporting than fiscal expansion in this model because its sales and interest-rate effects work in the same direction.

9.4 Policy comparison table

The table assumes fixed prices, lump-sum T, upward-sloping LM, and stable goods-market responses. Each row is a separate experiment. “Ambiguous” refers to general investment I(Y, i).

Policy	Curve shift	Y	i	C	I	Private S	Public T − G
T rises	IS left	↓	↓	↓	Ambiguous	↓	↑
T falls	IS right	↑	↑	↑	Ambiguous	↑	↓
G rises	IS right	↑	↑	↑	Ambiguous	↑	↓
G falls	IS left	↓	↓	↓	Ambiguous	↓	↑
M rises	LM down	↑	↓	↑	↑	↑	Same
M falls	LM up	↓	↑	↓	↓	↓	Same

A change in c_0 deserves separate saving analysis because it enters S directly. For example, lower c_0 shifts IS left, lowering Y and i, but the effect on private saving is not obtained simply by copying the tax-increase row. Always return to $S = -c_0 + (1 - c_1)(Y - T)$.

9.5 Combining fiscal and monetary policies

A fiscal contraction shifts IS left and a monetary expansion shifts LM down. Both lower i, but their output effects oppose each other. Depending on their sizes, Y can fall, remain unchanged, or rise.

The lecture uses the Clinton-era policy discussion to illustrate the possibility of monetary easing offsetting the demand loss from fiscal consolidation. If output is held constant while T rises or G falls, i must fall enough to stimulate the required extra investment.

If consolidation is through taxes and Y stays constant, consumption falls because Y − T falls. Investment must rise to fill that gap. If consolidation is through lower G with T and Y unchanged, consumption is unchanged and investment rises to offset lower government purchases. The same total output can therefore have a different expenditure composition.

Policy combination	Robust direction under the standard assumptions	Outcome depending on relative size
Fiscal expansion + monetary expansion	Y rises	i
Fiscal contraction + monetary expansion	i falls	Y
Fiscal expansion + monetary contraction	i rises	Y
Fiscal contraction + monetary contraction	Y falls	i

Near the lower bound, the central bank may be unable to supply the interest-rate reduction needed to fully offset a fiscal contraction through the model's conventional channel.

9.6 Additional practice: solve the entire model numerically

The lecture supplies general functional relationships rather than this particular linear system. This added example makes all the comparative statics calculable. Its additive real-money-demand function is a convenient alternative specification; it is **not an exact algebraic rewriting** of YL(i).

Assume:

C = 100 + 0.6(Y − T).
I = 150 + 0.1Y − 1000i.
M/P = 0.5Y − 2000i.
Initially T = 100, G = 100, and M/P = 300.

Goods market: Y = 100 + 0.6(Y − 100) + 150 + 0.1Y − 1000i + 100, so **0.3Y = 290 − 1000i**.

Money market: 300 = 0.5Y − 2000i, so i = 0.00025Y − 0.15.

Substitute: 0.3Y = 290 − 1000(0.00025Y − 0.15) = 440 − 0.25Y. Hence 0.55Y = 440, giving **Y = 800 and i = 0.05**.

Then C = 100 + 0.6 × 700 = **520**, I = 150 + 80 − 50 = **180**, private saving = 800 − 100 − 520 = **180**, and public saving = 0. Check: 520 + 180 + 100 = 800 and 0.5 × 800 − 2000 × 0.05 = 300.

General solution for this linear class

For $C = c_0 + c_1(Y - T)$, $I = I_0 + aY - bi$, and real-money demand $kY - hi$, define $m = M/P$, $A = c_0 - c_1T + I_0 + G$, and $\alpha = 1 - c_1 - a$. Then:

Equations: $Y = [A + (b/h)m]/[\alpha + bk/h]$.
$i = (kY - m)/h$.

The spending multiplier is $1/[\alpha + bk/h]$, and the tax multiplier is $-c_1/[\alpha + bk/h]$. The money multiplier with respect to **real money balances** is $(b/h)/[\alpha + bk/h]$. For nominal M at fixed P, multiply that last expression by 1/P.

The term bk/h captures the financial-market feedback: higher income raises money demand and i, which dampens investment. At a fixed interest rate the spending multiplier is 1/α instead.

For equal increases in G and lump-sum T, this IS-LM model gives a multiplier of $(1 - c_1)/[\alpha + bk/h]$, which need not equal 1. In the numerical example it is 0.4/0.55 ≈ 0.7273. The balanced-budget multiplier of 1 belongs to the earlier fixed-investment model.

Separate policy experiments from the same baseline

Scenario	Y	i	C	I	G	Private S	Public S
Baseline	800	5%	520	180	100	180	0
G rises by 55	900	7.5%	580	165	155	220	−55
Real money m rises by 110	900	2%	580	220	100	220	0
T rises by 50 and m rises by 60	800	2%	490	210	100	160	50

In the fiscal expansion, the sales effect on investment is +0.1 × 100 = +10, while the rate effect is −1000 × 0.025 = −25. Net investment falls by 15. This particular parameter choice produces crowding out in the level of investment; the general model allows other outcomes.

The monetary expansion achieves the same 100 output increase but lowers i. Investment rises by 10 from sales and 30 from the lower rate, for a total increase of 40.

In the policy mix, higher taxes reduce consumption by 30 at unchanged output. A three-percentage-point rate cut raises investment by 30. Public saving rises by 50 and private saving falls by 20, so national saving and investment both rise by 30. Every row satisfies $I = S + T - G$.

Deriving the offset needed to keep output constant

At a fixed Y in this linear model, $0 = \Delta G - c_1\Delta T - b\Delta i$. Thus $\Delta i = (\Delta G - c_1\Delta T)/b$. The money-market condition gives $\Delta m = -h\Delta i$.

For $\Delta T = 50$ and $\Delta G = 0$, $\Delta i = -0.6 \times 50/1000 = -0.03$. Required $\Delta m = -2000 \times (-0.03) =$ **60**. This is why the final row returns output to 800.

9.7 A repeatable answer structure

1. **State the model and what is held fixed.** For example, fixed P and M, lump-sum taxes, investment increasing in Y and decreasing in i.
2. **Identify the initial mechanism.** A tax rise lowers disposable income and consumption at each Y.
3. **Name the curve shift.** IS shifts left; LM is unchanged.
4. **Explain the feedback and final signs.** Lower Y reduces money demand, so i falls; both Y and i end below their original levels.
5. **Determine other variables from equations.** C falls; public saving rises; investment is ambiguous because its two determinants change in opposite directions.
6. **Check consistency.** Goods and money markets must both clear; saving and investment must agree under the relevant accounting convention.

This structure also prevents an unjustified claim that fiscal policy “shifts LM” merely because the resulting output change shifts money demand in the underlying money-market diagram.

10. Revision sheet and self-test

10.1 Essential formulas with conditions

Topic	Formula	Condition or interpretation
GDP accounting	$Y = C + I + G + X - IM$	I includes inventory investment
Value added	Output value − intermediate inputs	Do not subtract wages again
Nominal GDP	$\Sigma p_t q_t$	Current prices and quantities
Fixed-base real GDP	$\Sigma p_b q_t$	Common base-year prices
Fisher growth	$\sqrt{(R^L R^P)} - 1$	Geometric average of quantity growth factors
Chained level	Previous chained level × F_t	Reference-year level equals nominal GDP
GDP deflator	Nominal GDP/real GDP	Base 1, or multiply by 100
Inflation	$P_t/P_{t-1} - 1$	Growth of the price index
Nominal growth	$(1 + g^R)(1 + \pi) - 1$	Exact relationship
Unemployment	$U/(N + U)$	Only labour-force members in denominator
Participation	$(N + U)$/working-age population	Different denominator from unemployment
Consumption	$c_0 + c_1(Y - T)$	$0 < c_1 < 1$
Simple output	$(c_0 + \bar{I} + G - c_1 T)/(1 - c_1)$	Fixed investment, lump-sum taxes
Spending multiplier	$1/(1 - c_1)$	Basic Keynesian cross
Tax multiplier	$-c_1/(1 - c_1)$	Change measured as a rise in T
Balanced-budget multiplier	1	Equal changes in G and lump-sum T; fixed I
Private saving	$-c_0 + (1 - c_1)(Y - T)$	Linear consumption model
National saving	$S + T - G = I$	Closed-economy equilibrium
Bond yield	$F/P^B - 1$	One-year zero-coupon bond
Bond price	$F/(1 + i)$	Rate in decimal form
Money demand	$M^d = PYL(i)$	$L'(i) < 0$
Velocity	$V = PY/M$	$MV = PY$ by definition of V
IS	$Y = C(Y - T) + I(Y, i) + G$	Goods-market equilibrium
LM	$M/P = YL(i)$	Money-market equilibrium

10.2 Graph checklist

Graph	Horizontal axis	Vertical axis	Key slope or shift
Consumption function	Disposable income Y^D	Consumption C	Slope c_1; intercept c_0
Keynesian cross	Output/income Y	Demand Z and production	ZZ slope c_1, or c_1 + MPI; 45° slope 1
Money market	M, or M/P consistently	Interest rate i	Demand downward; fixed supply vertical
Bond market in Tutorial 3	Bond quantity	Interest rate i	Demand upward because a higher i means a lower price
IS-LM	Real output Y	Interest rate i	IS downward; course LM upward

Label the original and new equilibria, write the exogenous shock beside the curve that shifts, and mark the final changes in Y and i. A change in an axis variable ordinarily moves along a fixed curve; a change in something held fixed when constructing that curve can shift it.

10.3 Common mistakes and their repairs

Mistake	Repair
Adding sales of both intermediate and final goods	Count final output or sum value added
Treating value added as profit	Wages are part of value added
Treating a new house as household consumption	New residential construction is investment
Counting a transfer as G	Trace its effect through net taxes and household spending
Forgetting negative inventories when old output is sold	Record the offsetting inventory withdrawal
Averaging two growth rates to form a Fisher link	Take the square root of the product of the two growth factors
Using 3,050 as the tutorial's chained year 1 level	Divide 3,366 by F_2, giving 3,045.78
Interpreting disinflation as falling prices	Prices still rise when inflation remains positive
Using total population in the unemployment denominator	Use N + U
Using $1/(1 - c_1)$ for every model	Check investment, taxes, and interest-rate assumptions
Giving a positive tax multiplier for ΔT	The multiplier on a tax increase is negative
Saying T rises but public saving is unchanged	At fixed G, T − G rises
Equating bond price with bond yield	They move in opposite directions for a fixed payoff
Saying Y rising shifts LM	At fixed M/P it causes movement along LM
Saying fiscal expansion always reduces total I	Compare the sales and interest-rate effects
Treating nominal M and real M/P as identical	Account for the price-level convention

10.4 Retrieval questions

Cover the answers below and explain each result in complete sentences before using formulas.

1. Why do production, income, and expenditure measure the same GDP?
2. Why is a firm's wage bill not subtracted from value added?
3. Can real GDP exceed nominal GDP? Give the relevant price comparison.
4. How does changing a chain series' reference year differ from changing fixed-base prices?
5. Why can unemployment fall without employment rising?
6. If $c_1 = 0.8$, what are the spending and tax multipliers in the basic Keynesian cross?
7. With that MPC, what happens to Y if G rises by 20 and T rises by 20?
8. Why does an economy-wide attempt to save more fail to increase saving in the fixed-investment model?
9. A bond pays 100 next year and costs 96 today. What is its yield?
10. If nominal income rises while money supply is fixed, what happens to i and bond prices?
11. Why is IS downward-sloping, while the course's LM is upward-sloping?
12. A government raises T. What happens to C, public saving, private saving, and I in IS-LM?
13. Why does monetary expansion unambiguously raise investment in the standard model?
14. What policy mix can reduce the budget deficit while keeping output unchanged?
15. Why is a zero-lower-bound limitation not the same assumption as an interest-rate target?

Answer key

1. Producing final output generates value added, that value added generates factor income, and the output is used as final expenditure or inventory investment. Consistent accounting links all three measures.

2. Value added is the income available to labour and capital after intermediate inputs are paid for. Subtracting wages would move toward profit instead.

3. Yes. For example, if the base-year prices used in real GDP exceed current prices, the constant-price value can exceed the nominal value. This is not a contradiction.

4. A reference-year change rescales a given chain series. A fixed-base price change alters the weights placed on different quantities and can change growth rates.

5. People can stop searching and leave both U and L. The unemployment rate can then fall while N is unchanged.

6. Spending multiplier = 1/0.2 = 5. Tax multiplier = −0.8/0.2 = −4.

7. $\Delta Y = 5 \times 20 - 4 \times 20 = 20$, under the basic model's fixed-investment and lump-sum-tax assumptions.

8. Lower consumption reduces demand and equilibrium income. With I, T, and G fixed, equilibrium private saving is fixed by $S = I - T + G$.

9. $i = 4/96 = 0.0416667$, approximately 4.1667%.

10. Money demand rises. The equilibrium interest rate rises and the bond price falls.

11. A higher i reduces investment and goods-market equilibrium Y. A higher Y raises money demand, requiring a higher i at fixed real money supply.

12. C falls, public saving rises, and private saving falls. Investment is ambiguous because both Y and i fall.

13. It raises Y and lowers i, so both determinants stimulate investment.

14. Fiscal contraction with sufficient monetary expansion. The required easing lowers i enough to offset the reduction in goods demand, if the lower bound does not prevent it.

15. A rate target is a policy choice supported by an accommodating money supply. A liquidity trap reflects the inability to reduce the rate further through the model's conventional money-for-bonds operation.

10.5 A compact bilingual glossary

English	中文	Meaning to retain
Value added	增加值	Output value less intermediate inputs
Disposable income	可支配收入	Income after net taxes
Autonomous expenditure	自主支出	Spending independent of current income in the model
Marginal propensity to consume	边际消费倾向	Extra consumption per extra unit of disposable income
Multiplier	乘数	Total equilibrium response divided by the initial exogenous change
Paradox of thrift	节俭悖论 / 储蓄悖论	Desired extra saving can reduce income without raising aggregate saving
Automatic stabiliser	自动稳定器	Existing rules that dampen economic fluctuations
Liquidity	流动性	Ease of using an asset for payments
Opportunity cost	机会成本	Return or benefit forgone by choosing another use
Open market operation	公开市场操作	Central-bank purchases or sales of bonds
Liquidity trap	流动性陷阱	Extra money need not reduce the interest rate at its lower bound
Fiscal consolidation	财政整顿 / 财政紧缩	Measures to reduce the budget deficit
Crowding out	挤出效应	A channel through which higher interest rates reduce private investment
Endogenous / exogenous	内生 / 外生	Determined inside / taken as given in the model
Equilibrium	均衡	A configuration satisfying the model's consistency conditions

Appendix A. Source and coverage map

The source labels used throughout refer to the files below. Slide numbers count the title slide as slide 1. Tutorial page numbers refer to PDF pages, including inserted handwritten pages, rather than the sometimes inconsistent footer numbering.

Label	Supplied file	Extent	Main coverage in these notes
L1	Slides_export/Lecture 1_Intro.pptx	40 slides	Chapter 1: motivation, growth versus cycles, course framework
L2-3	Slides_export/Lecture 2_3_Measurement.pptx	50 slides	Chapters 2-4: GDP, chain indexes, labour statistics, price indexes
L4-5	Slides_export/Lecture 4_5_GoodsMarket.pptx	56 slides	Chapters 2, 5-6: expenditure, consumption, equilibrium, saving, fiscal policy
L6-7	Slides_export/Lecture 6_7_FinancialMarkets.pptx	38 slides	Chapter 7: money, bonds, monetary policy, liquidity trap
L8-9	Slides_export/Lecture 8_9_IS_LM.pptx	51 slides	Chapters 8-9: IS, LM, joint equilibrium, policy feedback and mixes
T1	Tutorial_export/26fall_T1_annotated.pdf	20 pages	Chapters 2-3: production examples, cars, full chain-index exercise
T2	Tutorial_export/26fall_T2 _annotated.pdf	18 pages	Chapters 1, 5: maths, Keynesian cross, numerical equilibrium, pandemic shock
T3	Tutorial_export/26fall_T3.pdf	15 pages	Chapter 7: money-market calculation, bond yields, all OMO prompts

Total source coverage: 235 lecture slides and 53 tutorial pages. The separate top-level Lecture 2_3_Measurement.pptx is identical to the copy in Slides_export and is counted once. The textbook, problem sets, and separate GDP spreadsheets are present in the folder but are outside this request's lecture-and-tutorial source set.

Assigned reading as listed in the decks: Blanchard Chapter 1 for the introduction; Chapter 2 except section 2-4 for measurement; Chapter 3 for the goods market; Chapter 4 except section 4.3 for financial markets; Chapter 5 for IS-LM, using the lecture's upward-sloping LM convention. The notes explain the supplied course materials without claiming coverage of textbook sections not developed in those materials.

Tutorial completion map: T1 pages 8-9, apples and juice, appears in section 2.2; pages 10-11, missing firm values, in section 2.2; page 15, cars, in section 3.4; pages 16-20, chain linking, in sections 3.2-3.3. T2 pages 3-8, maths, appears in section 1.6; pages 13-15, equilibrium, in section 5.5; pages 16-18, pandemic, in section 5.6. T3 pages 3-5, money demand and supply, appears in sections 7.1-7.4; page 6, the image-based calculation, in section 7.4; pages 7-9 and 12-15, operations and bond-market graphs, in section 7.5; pages 10-11, yields, in section 7.2.

Appendix B. Corrections and interpretation notes

These corrections preserve the course's intended models while repairing arithmetic, notation, or overly broad wording. They should be used when comparing the notes with the original slides.

Location	Issue	Correct interpretation
L4-5, slides 46-47	A tax arrow points down in the balanced-budget example	Both G and T rise by equal amounts; $\Delta Y = \Delta G$ in the basic model
L8-9, slide 36	Public saving labelled unchanged after T rises	$T - G$ rises when G is fixed; the speaker notes also indicate an increase
T3, page 15	A line says “To increases money supply” within the contractionary example	Central-bank bond sales decrease money supply
T1, page 20	Chained year 1 GDP displayed as approximately 3,045	Full-precision calculation gives 3,045.7802, or 3,045.78 to two decimals
L6-7, slides 18-19	Price 99 gives a yield rounded to 1%	Exact yield on face value 100 is 1/99 = 1.0101%
L6-7, slide 17	Principal shown with a percent sign before 5,000	Principal repayment is 5,000 currency units, in addition to the final coupon
L4-5, slide 5	Imports briefly labelled I	Use IM for imports and reserve I for investment
L2-3, slide 29	Chain-growth notation has ambiguous time subscripts	Multiply adjacent-year factors (1 + g) forward; divide backward
T1, pages 8-9	Same firm example has inconsistent year labels	Data and GDP result are the same; GDP = 2,600
L2-3, slide 43; T1, page 15	Same car data use different calendar years	Use the middle year as base; calculations are identical
T1, page 6	Final goods described as directly consumed	Final uses also include investment, such as new machines and houses
T1, page 12; L2-3, slide 31	Broad statements about non-cash or unreported activity	Distinguish the production boundary from measurement limitations and imputation
L4-5, slide 14	Small net exports motivate NX = 0	This is a simplifying assumption; small net trade does not imply small gross trade
L2-3, slide 42	Index level described as having no economic interpretation	Its absolute scale is arbitrary, but comparison with its reference level is meaningful

What was added for understanding: the labour-force numerical example; the exact nominal-growth decomposition; the algebraic automatic-tax stabiliser; explicit linear IS and IS-LM derivations; numerical policy comparisons; and the retrieval questions. These additions follow the stated assumptions and are distinguished from the original tutorial questions.

Final study principle: Before using a formula, state the model's assumptions. Before giving a policy conclusion, trace the initial effect and the feedback. Before accepting a numerical result, check it against the accounting identity and equilibrium conditions.